

INVERCARGILL, New Zealand

WMO: 938450

Lat: 46.4108S

Lon: 168.3178E

Elev: 9

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	30.4	26.5	20.4	29.1	28.5	22.5	31.8	32.4	45.0	29.3	44.7	3.7	30	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.5	74.6	62.7	71.4	61.5	68.5	60.0	64.6	71.9	62.7	69.5	61.0	67.0	11.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	81.4	68.1	59.6	76.2	65.9	58.1	72.2	64.1	29.7	71.9	28.3	69.5	27.1	67.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
33.2	29.5	26.0	DB	24.8	81.4	1.9	3.0	23.4	83.6	22.3	85.3	21.2	87.0	19.8	89.2
			WB	24.4	68.0	2.0	1.8	22.9	69.3	21.8	70.3	20.7	71.3	19.3	72.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	57.2	57.2	54.9	51.1	47.3	43.3	42.5	44.2	47.3	49.5	52.0	55.6
	DBStd	6.81	4.72	4.77	4.70	4.52	4.54	4.28	4.30	4.02	4.30	4.34	4.77	4.78
	HDD50	978	1	1	7	39	109	203	234	184	103	63	28	6
	HDD65	5440	247	224	313	417	550	650	697	645	532	481	390	294
	CDD50	1028	224	203	160	72	24	3	2	4	21	47	89	179
	CDD65	13	5	5	1	0	0	0	0	0	0	0	0	2
	CDH74	119	47	36	9	0	0	0	0	0	0	0	3	24
	CDH80	11	6	4	0	0	0	0	0	0	0	0	0	1
Wind	WSAvg	10.6	12.4	10.7	10.4	9.9	9.5	9.2	8.4	8.7	10.7	12.5	12.6	11.7
Precipitation	PrecAvg	42.8	4.2	3.1	3.7	3.8	4.4	4.0	3.2	2.8	3.2	3.5	3.5	3.8
	PrecMax	56.9	8.5	6.5	8.3	7.6	7.6	7.9	6.8	6.4	6.8	6.4	6.4	9.1
	PrecMin	32.0	1.5	0.5	1.1	0.6	1.8	1.1	1.1	0.6	0.4	1.2	0.9	1.3
	PrecStd	5.4	1.9	1.5	1.7	1.9	1.4	1.5	1.3	1.2	1.5	1.3	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.7	78.5	75.1	68.7	63.8	57.6	56.6	59.4	65.1	69.0	72.5	77.3
		MCWB	65.8	65.4	62.6	59.1	56.0	51.2	50.0	51.2	54.3	57.9	60.9	63.6
	2%	DB	73.5	73.7	70.2	64.7	59.7	54.0	53.6	55.7	61.0	64.5	68.0	72.5
		MCWB	62.6	63.2	61.4	56.7	53.7	49.4	48.2	49.4	52.3	55.5	59.0	61.3
	5%	DB	69.9	70.0	67.1	61.9	57.1	52.1	51.3	53.5	57.9	61.0	64.5	68.3
		MCWB	60.6	61.4	59.7	55.2	52.4	48.7	47.2	48.3	50.9	53.6	56.7	59.4
	10%	DB	66.6	66.5	63.9	59.1	54.8	50.5	49.3	51.4	55.3	58.0	60.9	64.9
		MCWB	59.3	59.8	58.1	54.0	51.1	47.7	46.1	47.2	49.6	52.0	54.3	57.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	67.5	65.4	60.5	57.2	52.3	51.1	52.0	55.6	59.1	62.7	65.2
		MCDB	76.1	76.2	71.9	66.5	61.4	55.9	55.3	57.5	62.9	66.7	69.6	74.5
	2%	WB	64.2	64.4	62.6	57.7	54.6	50.4	49.1	50.2	53.2	56.2	59.6	62.4
		MCDB	71.2	71.9	68.2	62.6	58.5	52.9	52.2	54.5	59.2	62.8	66.5	70.2
	5%	WB	62.0	62.2	60.5	56.0	52.9	49.2	47.7	48.9	51.4	54.3	57.3	60.4
		MCDB	68.5	68.7	65.8	60.3	56.3	51.6	50.4	52.6	56.6	60.0	63.0	66.6
	10%	WB	59.9	60.3	58.5	54.6	51.4	47.8	46.5	47.7	50.1	52.5	55.3	58.4
		MCDB	65.4	65.6	63.3	58.4	54.4	50.3	49.0	50.9	54.4	57.3	60.1	64.1
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.7	14.6	13.9	12.5	11.7	12.3	13.5	13.7	13.5	13.5	14.3
		MCDBR	22.0	21.4	20.9	18.9	16.1	13.6	14.0	16.3	18.7	19.5	20.2	21.3
		MCWBR	12.9	12.8	13.1	12.3	11.7	10.7	10.7	11.6	12.1	12.5	12.6	12.2
	5% WB	MCDBR	19.5	19.3	18.2	15.9	14.5	12.3	12.4	14.4	16.4	17.8	18.4	19.2
		MCWBR	12.5	12.2	12.1	11.1	11.2	10.4	10.1	10.8	11.2	12.0	11.8	12.1
Clear-Sky Solar Irradiance	taub	0.343	0.330	0.317	0.309	0.300	0.300	0.291	0.301	0.325	0.327	0.330	0.330	
	taud	2.501	2.555	2.593	2.597	2.584	2.532	2.579	2.580	2.501	2.511	2.515	2.545	
	Ebn at Noon	309	304	292	270	245	228	245	268	284	302	312	316	
	Edh at Noon	35	31	27	22	18	17	18	22	29	32	34	34	
All-Sky Solar Radiation	RadAvg	1827	1583	1151	733	430	328	401	639	1009	1427	1752	1942	
	RadStd	137	106	56	46	28	26	27	37	59	91	115	123	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.17	N/A	N/A	N/A	-95	-238	+137	N/A
Regional (6 neighbors)	N/A	N/A	+1.26	N/A	N/A	N/A	-85	-202	N/A	N/A

Nomenclature: See separate page